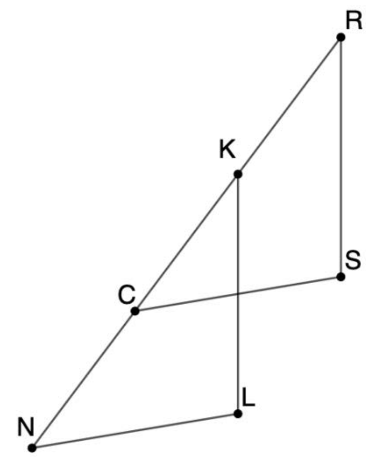
Geometry
Unit 6
Lesson 9 Practice

Name____________________
Date____________________
Period__________________

Given: $\overline{NL} \parallel \overline{CS}$ and $\angle KRS$ and $\angle LKR$ are supplementary

Prove: $\triangle NKL \sim \triangle CRS$



Complete the two-column proof with the provided statements and reasons.

	Statement	Reason
	$\overline{NL} \parallel \overline{CS}$	Given
1.		
2.		
3.		
4.		
5.		

$\triangle NKL \sim \triangle CRS$ — AA ~

$\angle KRS$ and $\angle LKR$ are supplementary angles — If two parallel lines are intersected by a transversal, then corresponding angles are congruent.

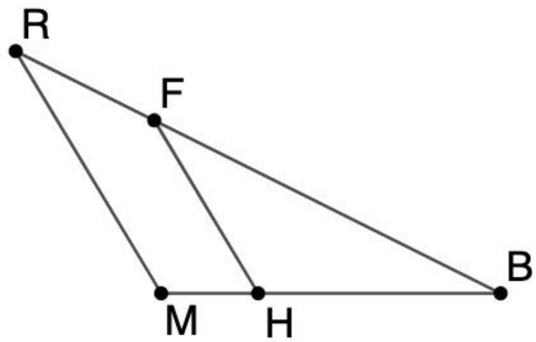
$\angle NKL \cong \angle CRS$ — If consecutive interior angles are supplementary, then two lines intersected by a transversal are parallel.

$\overline{KL} \parallel \overline{RS}$ — Reflexive Property of Congruence

$\angle KNL \cong \angle RCS$ — Given

Given: $RF = \frac{2}{7}RB$ and $MH = \frac{2}{7}MB$

Prove: $\triangle RBM \sim \triangle FBH$



Complete the two-column proof with the provided statements and reasons.

	Statement	Reason
6.		
7.		
8.		
9.		
10.		

$\frac{RF}{RB} = \frac{2}{7}$ and $\frac{MH}{MB} = \frac{2}{7}$	SAS ~
$\triangle RBM \sim \triangle FBH$	Given
$\frac{RF}{RB} = \frac{MH}{MB}$	Reflexive Property of Congruence
$RF = \frac{2}{7}RB$ and $MH = \frac{2}{7}MB$	Substitution Property of Equality
$\angle RBM \cong \angle FBH$	Division Property of Equality